

Saddle-node bifurcation of first-passage-time densities induced by a directed shortcut

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First-passage-time (FPT) densities in systems that possess a shortcut to the target are known to be bimodal: a fast “direct” peak and a slow “indirect” peak. We show that this bimodality is the observable-level signature of a *saddle-node bifurcation* that can be solved exactly. Modeling a directed shortcut to an absorbing target as a rank-one killing defect on a lazy lattice—whose diffusive limit is Brownian motion on an interval with an interior δ -sink of strength b at fractional position θ —we show that the second FPT peak is born and annihilated at a fold of a *signed* spectral mixture, governed by the compact criterion $S_1 = S_2 = 0$, where $S_n(\tau) = \sum_j G_j \mu_j^n e^{-\mu_j \tau}$ and $f' = -S_1$, $f'' = S_2$. The double peak exists if and only if $0 < b < b_c(\theta)$, with an explicit boundary $b_c(\theta)$ that is symmetric about $\theta = \frac{1}{2}$, has a minimum near $\theta \simeq 0.381$, and diverges as $b_c \sim 0.789/\min(\theta, 1 - \theta)$ near the endpoints; at least three sign-alternating modes are required, and the fold obeys the universal normal-form scaling $\tau_+ - \tau_- \propto (b_c - b)^{1/2}$, $\Delta f \propto (b_c - b)^{3/2}$. The mechanism is generic rather than lattice-specific: it follows model-independently from a low-rank non-Hermitian defect, it generalizes to multiple shortcuts (two shortcuts create a *third* peak organized by a cusp catastrophe, which we locate), and it survives on a two-dimensional lattice. The threshold b_c is a time-domain morphology fold, *not* a spectral exceptional point. The result recasts a known phenomenology as a predictive, exactly solvable bifurcation with a measurable threshold and universal scaling, testable in feedback-controlled colloidal search and in engineered transport networks.

I. INTRODUCTION

The first-passage time (FPT) to a target is a central observable of stochastic transport, search and reaction kinetics [1–3]. When a searcher has access to a “shortcut”—a relocation channel, a long-range rewired edge, or an active-transport route—the FPT density $f(t)$ can become bimodal: fast trajectories that use the shortcut form an early peak, while slow trajectories that explore the domain form a later peak, provided the two time scales are well separated [6, 8]. Multimodal capture-time densities are likewise known for interior partially absorbing traps when several traps are strategically arranged [9, 10], and exact spatiotemporal solutions of confined lattice random walks with defects are by now a mature program [4, 5].

These works *exhibit* multimodality; they do not, however, classify or predict *when* a second peak exists as a control parameter is varied, nor identify the transition as a bifurcation with a sharp, computable threshold. Here we supply exactly that. For a minimal directed shortcut—modeled as a rank-one killing (non-Hermitian) defect—we prove that the birth and death of the second FPT peak is a *saddle-node* of the density’s critical points, give the exact existence boundary $b_c(\theta)$, a minimal-mode theorem, the universal fold scaling, and demonstrate that the mechanism is generic (multiple shortcuts, two dimensions). The broad message is that FPT distributions carry controllable time-domain structure—an emergent arrival-time scale with a threshold—that is invisible to the mean first-passage time and to spectral gaps.

We are careful to separate what is new from what is not. Shortcut-induced bimodality and interior-trap multimodality are known [6, 9, 10]; the lattice Green-function/defect method is Giuggioli’s [4, 5]; and Ref. [7] concerns bimodality of a splitting probability $P(\omega)$, not of $f(t)$. Our contribution is the exactly solvable *bifurcation-theoretic* account: the fold criterion, the predictive threshold, and the universality.

II. MODEL AND EXACT SOLUTION

Consider a lazy random walk on a ring of N sites: at each step the walker stays with probability $1 - q$ and hops to each neighbor with probability $q/2$. Site $v = 0$ is an absorbing target. A directed shortcut at a source site u diverts probability $\lambda = \beta(1 - q)$ from the u self-loop onto the directed edge $u \rightarrow v$. Removing the absorbed state leaves a transient generator on sites $1, \dots, N - 1$ that is a Dirichlet path with a rank-one diagonal killing defect at u ,

$$Q_b = Q_0 - b |u\rangle\langle u|, \quad (1)$$

so probability leaks irreversibly from u into the target. Although the shortcut is directed, the transient block remains symmetric (the directedness enters only the absorption vector), so its spectrum is real.

The generating function is rational with an exact Montroll-type determinant. With U_m the Chebyshev polynomials of the second kind, $y = 1 + (z^{-1} - 1)/q$, and $a = q/\lambda$, the characteristic determinant for a source at u is

$$D_u(y) = a U_{N-1}(y) + 2 U_{u-1}(y) U_{N-u-1}(y), \quad (2)$$

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and the FPT probability is $F(t) = \sum_j B_j s_j^{t-1}$ with $s_j = 1 - q + qy_j$, $D_u(y_j) = 0$, and residues $B_j = q\mathcal{N}_{r,u}(y_j)/D'_u(y_j)$ (Appendix A). The antipodal case $u = N/2$ reduces to $D \propto aT_L + U_{L-1}$, recovering the symmetric problem as a special case of the rank-one theory.

Diffusive limit. Writing $\theta = u/N$, $b = \beta(1-q)N/q$, $\tau = qt/N^2$ and $s = 1 - q + q \cos(2w/N)$, the walk converges to Brownian motion on $[0, 1]$ with absorbing ends and an interior δ -sink,

$$\partial_\tau p = \frac{1}{2} p_{xx} - b \delta(x - \theta) p, \quad p(0) = p(1) = 0. \quad (3)$$

Its spectrum is fixed by

$$D_\theta(k; b) = k \sin k + 2b \sin(k\theta) \sin(k(1 - \theta)) = 0, \quad (4)$$

with $k = 2w$ and $\mu_j = k_j^2/2 = 2w_j^2$; for $\theta = \frac{1}{2}$ this is $\tan w = -2w/b$. The FPT density is $\Phi_{\xi,\theta}(\tau; b) = \sum_j G_j e^{-\mu_j \tau}$, started at $\xi = r/N$. The *signed* amplitudes are (Appendix B)

$$G_{\xi,\theta}(w) = 2w^2 \frac{\phi_{w,\theta}(\xi) I(w, \theta)}{J(w, \theta)}, \quad (5)$$

where $\phi_{k,\theta}(x) = \sin(k(1 - \theta)) \sin(kx)$ for $x \leq \theta$ and $\sin(k\theta) \sin(k(1 - x))$ for $x \geq \theta$, and $I = \int_0^1 \phi dx$, $J = \int_0^1 \phi^2 dx$ (closed forms in Appendix B). Modes with $\sin(n\pi\theta) = 0$ do not see the sink and contribute the free term $G_n^{\text{mode}} = n\pi[1 - (-1)^n] \sin(n\pi\xi)$. We have verified (5) and the full master curve against exact finite- N residues to a relative error $\sim 10^{-5}$ with clean $O(N^{-2})$ convergence.

III. THE SADDLE-NODE EXISTENCE THEOREM

Define the signed spectral moments

$$S_n(\tau; b) = \sum_j G_j \mu_j^n e^{-\mu_j \tau}, \quad (6)$$

so that $\Phi' = -S_1$ and $\Phi'' = S_2$. An interior extremum of the FPT density is a zero of S_1 ; two such extrema (a valley-peak pair, i.e. the second peak) merge and annihilate at a *saddle-node* fold,

$$\boxed{S_1(\tau_c, b_c) = 0, \quad S_2(\tau_c, b_c) = 0}, \quad (7)$$

with nondegeneracy $S_3(\tau_c, b_c) \neq 0$ and $\partial_b S_1(\tau_c, b_c) \neq 0$. No positivity is assumed—the G_j are signed—so this is the correct criterion for a fold of a signed exponential mixture. Consequently the second peak exists *iff* $0 < b < b_c(\theta)$; we verify numerically that b_c is a *single* fold, i.e. the double-peak set is a single connected interval in b with no re-entrant window.

Boundary. $b_c(\theta)$ is symmetric, $b_c(\theta) = b_c(1 - \theta)$, with a minimum at $\theta \simeq 0.381$ ($b_c \simeq 2.16$) and the endpoint law

$$b_c(\theta) \sim \frac{0.7890261736}{\min(\theta, 1 - \theta)}, \quad \tau_c \sim 0.1579221011 d^2, \quad (8)$$

$d = \min(\theta, 1 - \theta)$, obtained from a half-line boundary-layer problem. In the antipodal case $b_c(\frac{1}{2}) = 3.0764323604$, with the near-antipodal expansion $b_c(\frac{1}{2} + \epsilon) = 3.0764323604 - 133.107 \epsilon^2 + O(\epsilon^4)$. Fig. 1(a).

Normal form. Near the fold the two extrema split as $\tau_+ - \tau_- \propto (b_c - b)^{1/2}$, $\Phi(\tau_+) - \Phi(\tau_-) \propto (b_c - b)^{3/2}$, (9)

the generic saddle-node signature; we confirm both exponents numerically [Fig. 1(c)], which certifies the transition as a fold rather than a spectral crossover.

Minimal mode count. A two-mode fold is impossible, even with signs: $S_1 = 0$ forces $A_2 = -A_1$ ($A_j = G_j \mu_j e^{-\mu_j \tau}$), whence $S_2 = (\mu_1 - \mu_2) A_1 = 0$ gives $A_1 = A_2 = 0$. Thus at least three sign-alternating modes are required; in the antipodal case $G_1 > 0, G_2 < 0, G_3 > 0, \dots$, and retaining K affected modes gives $b_c^{(3)} = 3.046$, $b_c^{(5)} = 3.0764$.

IV. UNIVERSALITY

The fold (7) is not special to the ring. For any absorbing Markov generator, a rank- m killing defect $Q_b = Q_0 - UBU^\top$ gives, by the Woodbury identity,

$$\det(sI - Q_b) = \det(sI - Q_0) \det[I_m + BW(s)], \quad W_{ab} = \langle u_a | (sI - Q_0)^{-1} | u_b \rangle, \quad (10)$$

and the FPT density is a signed spectral mixture whose second-peak birth/death is the fold $S_1 = S_2 = 0$. The double-peak region is an *open* set (structurally stable by the implicit-function theorem), hence not fine-tuned. The honest universal statement is not “every shortcut creates a peak” but “whenever a shortcut-induced peak is born or annihilated generically, the mechanism is a fold of the signed spectral mixture.”

Multiple shortcuts. For two shortcuts at $\theta_1 = \alpha < \theta_2 = \beta$ with strengths b_1, b_2 , the secular determinant is

$$\begin{aligned} D_{12}(k) = & k \sin k + 2b_1 \sin(k\alpha) \sin(k(1 - \alpha)) \\ & + 2b_2 \sin(k\beta) \sin(k(1 - \beta)) \\ & + \frac{4b_1 b_2}{k} \sin(k\alpha) \sin(k(\beta - \alpha)) \sin(k(1 - \beta)), \end{aligned} \quad (11)$$

the last term being a genuine two-defect interaction (it vanishes as the sources coalesce, giving $b = b_1 + b_2$). Two shortcuts can produce a *third* FPT peak: at $(\alpha, \beta, b_1, b_2) = (0.38, 0.48, 1.35, 0.14)$, $\xi = 0.463$, the exact ring shows three peaks at $\tau \simeq 3.0 \times 10^{-4}, 5.3 \times 10^{-3}, 6.1 \times 10^{-2}$ [Fig. 1(d)]. The triple-peak wedge is organized by a *cusp* $S_1 = S_2 = S_3 = 0$; we locate it at

$$(\alpha, \beta, b_1, b_2, \tau_c) = (0.38, 0.489580512568, 1.35, 1.08389, 0.0012366), \quad (12)$$

with $S_4 \neq 0$ and $\det \partial(S_1, S_2)/\partial(b_1, b_2) = -6.9 \times 10^7 \neq 0$, a genuine codimension-two cusp (Appendix C). We note that the cusp sits slightly off the triple-peak example ($\beta \approx 0.48958$, not 0.48, which carries only ordinary folds).

Two dimensions. On an $L \times L$ torus with an absorbing target and a directed shortcut, the same algebra holds [Eq. (10) with the lattice resolvent]. Numerically, a single diffusive peak at $\beta = 0$ acquires a shortcut-induced capture peak; the two coexist over a window and the diffusive peak's prominence falls to zero at $\beta_c^{2D} \simeq 0.55\text{--}0.65$, a fold present at $L = 17, 25, 33$ [Fig. 1(e,f)]. The mechanism is therefore not one-dimensional. A single-site sink is marginal in 2D (its self-Green function is logarithmically divergent), so the bare $b_c(L)$ drifts as $\sim 1/\log L$; the robust, size-independent diagnostics are the moment condition (7) and the fold scaling (9).

Robustness. The fold persists under any perturbation that preserves $S_3 \neq 0$, $\partial_b S_1 \neq 0$: weak hop-rate disorder, small shifts of the source or start, and the laziness q (which merely rescales time, $\tau_c \rightarrow \tau_c/q$). It is destroyed only by placing the source on a symmetry node (dark modes), by source coalescence, or by pushing the start onto the shortcut (the early peak then escapes to $\tau = 0$, outside the interior-fold theorem).

V. PHYSICAL REALIZATION AND NON-HERMITIAN CHARACTER

Physically the shortcut is a localized target-delivery channel: while at u the searcher is delivered to the target at rate b , i.e. $f(t) = f_0(t) + b p_u(t)$, so a finite gate of width a and local trigger rate κ becomes a δ -sink of strength $\sim \kappa a$. The sharpest realization is feedback-controlled colloidal or active-particle search: a narrow gate at $x = \theta L$ (optical tweezers, a microfluidic pulse, or active-particle feedback) that, upon entry, occasionally delivers the particle to the target and ends the trial; b is the trigger rate, tunable per frame. Repeating many trials and scanning b at fixed θ (best near $\theta \simeq 0.38$, where b_c is smallest) should reveal the second peak for $b < b_c(\theta)$ and its annihilation at b_c . We caution that standard resetting-to-origin is *not* this problem—it reinjects probability into the survival domain; the correct protocol is delivery-and-terminate. A second realization is an engineered or small-world transport network with a one-way express edge to an absorbing node; two such edges provide a direct test of the cusp. Intracellular motor capture to a reaction site realizes it in a coarse-grained (ideally reconstituted) setting.

We stress what the non-Hermitian character is and is not. Q_b is a passive open-Markov loss defect, analogous to an effective non-Hermitian Hamiltonian with a localized absorber; it is *not* PT-symmetric (no balanced gain and loss). Crucially, b_c is *not* an exceptional point: an EP requires eigenvalue coalescence with a defective (Jordan) block, whereas here the eigenvalues remain real, simple

and smooth through b_c (verified), and the fold arises from cancellation among at least three signed residues. The $(b_c - b)^{1/2}$ scaling in (9) is a saddle-node of the scalar $f(t; b)$, not EP eigenvalue splitting—the same exponent for a different object. In the reversible diffusive limit the δ -sink operator is self-adjoint (Sturm–Liouville), so no spectral EP exists.

VI. DISCUSSION

We have shown that shortcut-induced FPT bimodality, long known phenomenologically, is the observable signature of an exactly solvable saddle-node bifurcation: a rank-one directed defect produces a second first-passage peak that exists precisely for $0 < b < b_c(\theta)$, is born/annihilated through a fold with universal $(1/2, 3/2)$ scaling, requires at least three sign-alternating modes, generalizes to a cusp for two defects, and survives in two dimensions. The single most defensible novelty is the bifurcation-theoretic classification with a predictive threshold, distinct from the known bimodality and from the underlying defect-resolvent method. The predicted fold exponents and the two-shortcut triple peak are directly measurable.

Beyond the results proved and verified here, the analysis leaves a small set of technical questions: a rigorous $N \rightarrow \infty$ statement requires, in addition to the standard convergence of fixed-mode residues (Appendix B), uniform C^1 convergence of the moment sums in a neighborhood of the fold; the large- L renormalized scaling of b_c in two dimensions; and whether the constants $0.7890\dots$ and the antipodal b_c admit elementary closed forms.

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Appendix A: Exact finite- N determinant and residues

Deleting the target from the lazy-ring transition matrix leaves the transient block $Q_0 - \lambda |u\rangle\langle u|$ on $\{1, \dots, N-1\}$. Using the matrix-determinant lemma on the rank-one defect and $\det[yI - \frac{1}{2}A_{\text{path}}] = 2^{-(N-1)}U_{N-1}(y)$, $[(yI - \frac{1}{2}A_{\text{path}})^{-1}]_{u,u} = 2U_{u-1}U_{N-u-1}/U_{N-1}$, one obtains (2). The piecewise numerator is $\mathcal{N}_{r,u} = a[A_r + A_{N-r}] + 2A_{N-u}[A_r + A_{u-r}]$ for $r \leq u$ (and the mirror for $r \geq u$), $A_m := U_{m-1}$. The channel (splitting) probability is $\pi_{\text{sc}}(r) = 2 \min(r, u)[N - \max(r, u)]/(aN + 2u(N-u))$, exact by Sherman–Morrison; for the antipodal case $\pi_{\text{sc}} = \rho/(a+L)$. We emphasize π_{sc} is the eventual splitting probability, not the first-peak mass.

Appendix B: Continuum amplitude

With $H_b = -\frac{1}{2}\partial_x^2 + b\delta_\theta$ and R_0 the sink-free interval resolvent, continued to $z = -k^2/2$, Sherman–Morrison gives $r_b = r_0 - b r_0(x, \theta) r_0(\theta, y) / [1 + b r_0(\theta, \theta)]$, whose poles are the zeros of (4); the residue eigenfunction is $\psi_j = \phi_{k_j, \theta}$, which satisfies the δ -sink jump $\psi'(\theta^+) - \psi'(\theta^-) = 2b\psi(\theta) \Leftrightarrow D_\theta = 0$ (verified to 2×10^{-14}). The absorption density $f = -\frac{d}{dt} \int p$ has Laplace transform $\tilde{f} = 1 - z\tilde{Q}$, and the residue at $z = -\mu_j$ yields $G_j = \mu_j \psi_j(\xi) \int \psi_j / \int \psi_j^2$, i.e. (5), with

$$I(k, \theta) = \frac{1}{k} [\sin(k(1 - \theta))(1 - \cos k\theta) + \sin(k\theta)(1 - \cos k(1 - \theta))], \quad (\text{B1})$$

$$J(k, \theta) = \frac{1}{2} [\theta B^2 + (1 - \theta) A^2] - \frac{AB \sin k}{2k}, \quad (\text{B2})$$

$A = \sin k\theta$, $B = \sin k(1 - \theta)$. The exact identity $J = \sin(k) D_k(k; b) / (4b)$ at a root (verified to 3×10^{-15}) links the norm to the determinant derivative and gives non-degeneracy of affected modes. For $\theta = \frac{1}{2}$, (5) reduces to $G_{\xi, 1/2} = 4w(1 - \cos w) \sin(2w\xi) / [\sin^2 w(1 + b(b+2)/4w^2)]$. Node modes are the free Dirichlet modes $\sin(n\pi x)$ at $\sin(n\pi\theta) = 0$, with the amplitude quoted in the text. The I, J closed forms match direct quadrature to 10^{-9} , and the reconstructed master curve matches exact finite- N residues to $\sim 10^{-5}$ with $O(N^{-2})$ convergence. Convergence of fixed-mode residues as $N \rightarrow \infty$ follows from norm-resolvent convergence of the killed generator; passing the fold (7) to the limit additionally requires uniform C^1 convergence of $S_{1,2,3}$ near (τ_c, b_c) , which we state as

a hypothesis and support by the observed $O(N^{-2})$ fold convergence.

Appendix C: Two-shortcut cusp

For two sources the rank-two eigenfunction is $\psi_j(x) = -\sum_a b_a c_a^{(j)} r_0(k_j; x, \theta_a)$ with $c^{(j)} \in \ker[I_2 + W(k_j)B]$, and $G_j = \mu_j \psi_j(\xi) \int \psi_j / \int \psi_j^2$ with $\int \psi_j = -\sum_a b_a c_a I_{\theta_a}$ and $\int \psi_j^2 = \sum_{ab} b_a c_a b_b c_b C_{ab}$; closed forms for I_θ and C_{ab} are elementary. The cusp solves $S_1 = S_2 = S_3 = 0$ with $S_4 \neq 0$ and a nonzero unfolding Jacobian $\det \partial(S_1, S_2) / \partial(b_1, b_2)$. At (12) an independent reimplementations gives normalized residuals $\tau S_1 / \Phi \approx 7 \times 10^{-12}$, $\tau^2 S_2 / \Phi \approx 5 \times 10^{-11}$, $\tau^3 S_3 / \Phi \approx 4 \times 10^{-10}$, $\tau^4 S_4 / \Phi \approx -1.67$, and $\det \partial(S_1, S_2) / \partial(b_1, b_2) \approx -6.9 \times 10^7$, confirming a nondegenerate cusp.

Appendix D: Numerical methods and reproducibility

Exact FPT densities use the symmetric transient block (`numpy eigvalsh`); high-precision checks use `mpmath`. Continuum spectra are the roots of (4)/(11); amplitudes use (5) and Appendix C. All scripts are deterministic; the figure is generated by `dpma_prr_figures.py`, the existence boundary by `dpma_saddle_node_bc.theta.py`, the amplitude by `dpma_general_u_master_curve.py`, the cusp by `dpma_cusp.verify.py`, and the 2D test by `dpma_2d.universality.py`.

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Directed-shortcut first-passage saddle-node: existence, universality (PRR figure suite)

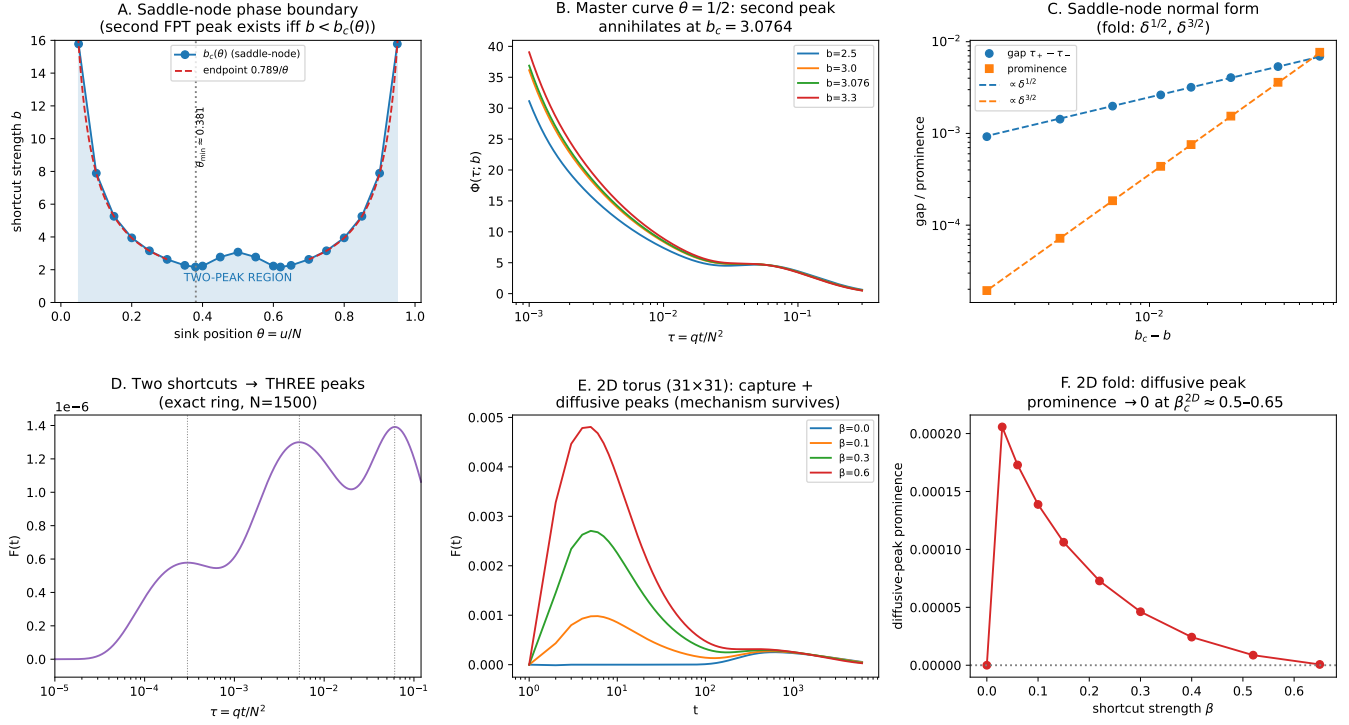


FIG. 1. (a) Saddle-node phase boundary $b_c(\theta)$: the second FPT peak exists for $0 < b < b_c(\theta)$; symmetric, minimum at $\theta \simeq 0.381$, with the endpoint law $0.789/\theta$ (dashed). (b) Master curve $\Phi(\tau; b)$ at $\theta = \frac{1}{2}$ as b crosses $b_c = 3.0764$. (c) Normal-form scaling: gap $\propto (b_c - b)^{1/2}$ and prominence $\propto (b_c - b)^{3/2}$. (d) Two shortcuts produce a third peak (exact ring, $N = 1500$). (e) 2D torus (31×31): capture and diffusive peaks. (f) 2D fold: the diffusive-peak prominence vanishes at $\beta_c^{2D} \simeq 0.55-0.65$.